

MULTIVARIATE ANALYSIS

多元分析

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1 Introduction

略。

2 Multivariate normal

2.1 Review of univariate normal distribution

The **univariate normal distribution**, with mean μ and variance σ^2 , has the probability density function (pdf)

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$$

denoted as $X \sim \mathcal{N}(\mu, \sigma^2)$.

Some theories:

- **Standardised:** If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$.
- **Chi-square distribution:**
 - If $Z \sim \mathcal{N}(0, 1)$, then $Z^2 \sim \chi^2(1)$;
 - If $Z_1, Z_2, \dots, Z_p \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$, then $Z_1^2 + Z_2^2 + \dots + Z_p^2 = \sum_{i=1}^p Z_i^2 \sim \chi_{(p)}^2$.

2.2 Density of multivariate normal

让我们说中文。

对于单一变量正态分布 pdf 的指数平方项可以写成如下形式

$$\frac{(x - \mu)^2}{\sigma^2} = (x - \mu)(\sigma^2)^{-1}(x - \mu).$$

类比地，将其从标量（scalar）写成向量或矩阵（vector/matrix）的形式，即

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}).$$

因此得到多元正态的 pdf.

定义 2.1 Multivariate normal (MVN)

A p -dimensional normal density for the random vector $X' = [X_1, X_2, \dots, X_p]$ has the form

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

We shall denote this p -dimensional normal density by $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

同时不难发现

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \sim \chi_{(p)}^2.$$

可以猜想到，一元正态分布是一个“钟形”分布，那么将其进行立体化，不难想象出二元正态分布应该是一个“山丘”的形状，因此该“山丘”的等高线必定是个椭圆。同理将该猜想一般化拓展到高维属性。

定义 2.2 Contour equation 等高线方程

常密度轮廓的 p 维正态分布是定义为 $\mathbf{x}$ 的椭球体，其方程为

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2 \sim \chi_{(p)}^2,$$

其中，椭球的中心 (center) 位于 $\boldsymbol{\mu}$ ，其两轴 (major/minor axis) 向量为 $\pm c\sqrt{\lambda_i} \mathbf{e}_i$ ，有

$$\boldsymbol{\Sigma} \mathbf{e}_i = \lambda_i \mathbf{e}_i,$$

即 λ_i 和 $\mathbf{e}_i$ 分别是 $\boldsymbol{\Sigma}$ 的特征值 (eigenvalue) 和特征向量 (eigenvector)，其中，两轴方向 (direction) 为 $\mathbf{e}_i$ ，两个半轴长度 (half-length) 为 $c\sqrt{\lambda_i}$ 。

定理 2.1

令 $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)^\top$ 为长度为 n 的向量分为两个子向量的长度 $(q, n - q)$ 。假设 $\mathbf{X}$ 服从多元高斯分布，即

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \right).$$

Then the conditional distribution of $\mathbf{X}_1 | \mathbf{X}_2$ is given by

$$\mathbf{X}_1 | \mathbf{X}_2 \sim \mathcal{N} (\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{X}_2 - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}).$$

很显然, $\mathbf{X}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$, $\mathbf{X}_2 \sim \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$, 并且 $\text{Cov}[\mathbf{X}_1, \mathbf{X}_2] = \boldsymbol{\Sigma}_{12} = \boldsymbol{\Sigma}_{21}^\top$. 因此

- $\mathbb{E}[\mathbf{X}_1 | \mathbf{X}_2] = \text{原均值} + \text{协方差占比} \times (\text{观测值} - \text{对应均值})$.
- $\text{Var}[\mathbf{X}_1 | \mathbf{X}_2] = \text{原方差} - \text{可被 } \mathbf{X}_2 \text{ 解释的部分}$.

2.3 Distribution of sample mean and variance

If $X_1, X_2, \dots, X_n$ are independently drawn from a $\mathcal{N}(\mu, \sigma^2)$ and $\bar{X} = \frac{1}{n}X_1 + \frac{1}{n}X_2 + \dots + \frac{1}{n}X_n$, then

$$\bar{X} \sim \mathcal{N} \left(\mu, \frac{\sigma^2}{n} \right).$$

定理 2.2 Distribution of sample mean

Using the properties of MVN, we can say if $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are independently drawn from $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then

$$\bar{\mathbf{X}} \sim \mathcal{N}_p \left(\boldsymbol{\mu}, \frac{1}{n} \boldsymbol{\Sigma} \right).$$

If $X_1, X_2, \dots, X_n$ are independently drawn from a $\mathcal{N}(\mu, \sigma^2)$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$,

then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2.$$

定义 2.3 Wishart distribution

The sampling distribution of the sample covariance matrix is called the **Wishart distribution**; it is defined as the sum of independent products of multivariate normal random vectors. Specifically,

$$\begin{aligned} \mathbf{W}_m(\cdot|\Sigma) &= \text{Wishart distribution with } m \text{ d.f.} \\ &= \text{distribution of } \sum_{i=1}^m Z_i Z_i^\top, \end{aligned}$$

where the Z_i are each independently distributed as $\mathcal{N}_p(\mathbf{0}, \Sigma)$.

性质 2.1

- 若 $\mathbf{A}_1 \sim \mathbf{W}_{m_1}(\mathbf{A}_1, \Sigma)$, 且与之独立的 $\mathbf{A}_2 \sim \mathbf{W}_{m_2}(\mathbf{A}_2, \Sigma)$, 则

$$\mathbf{A}_1 + \mathbf{A}_2 \sim \mathbf{W}_{m_1+m_2}(\mathbf{A}_1 + \mathbf{A}_2, \Sigma).$$

- 若 $\mathbf{A} \sim \mathbf{W}_m(\mathbf{A}, \Sigma)$, 则 $\mathbf{C}\mathbf{A}\mathbf{C}^\top \sim \mathbf{W}_m(\mathbf{C}\mathbf{A}\mathbf{C}^\top, \mathbf{C}\Sigma\mathbf{C}^\top)$.
- 若 $\mathbf{A} \sim \mathbf{W}_m(\mathbf{A}, \mathbf{I})$, 则 $\text{trace}(\mathbf{A}) \sim \chi_{(mp)}^2$.

2.4 Large sample properties

Recall: If $X_1, X_2, \dots, X_n$ are independently drawn from a $\mathcal{N}(\mu, \sigma^2)$ and $\bar{X} = \frac{1}{n}X_1 + \frac{1}{n}X_2 + \dots + \frac{1}{n}X_n$, then

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \Rightarrow \sqrt{n}(\bar{X} - \mu) \sim \mathcal{N}(0, \sigma^2) \Rightarrow \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1).$$

定理 2.3 The central limit theorem

Let $X_1, X_2, \dots, X_n$ be independent observations from any population with mean μ and finite covariance Σ . Then

$$\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \approx \mathcal{N}_p(\mathbf{0}, \Sigma) \Rightarrow \sqrt{n}\Sigma^{-\frac{1}{2}}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \approx \mathcal{N}_p(\mathbf{0}, \mathbf{I}),$$

and

$$n(\bar{\mathbf{X}} - \boldsymbol{\mu})\mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \approx \chi_{(p)}^2$$

3 Inference

3.1 Three major distributions based on $\mathcal{N}(0, 1)$

If $X_1, X_2, \dots, X_n$ are independently drawn from a $\mathcal{N}(\mu, \sigma^2)$ then

$$\begin{aligned}\bar{X} &\sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \\ \bar{X} - \mu &\sim \mathcal{N}\left(0, \frac{\sigma^2}{n}\right) \\ \sqrt{n}(\bar{X} - \mu) &\sim \mathcal{N}(0, \sigma^2)\end{aligned}$$

$$\boxed{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)}$$

3.1.1 χ^2 distribution

定义 3.1 χ^2 distribution

If $Z \sim \mathcal{N}(0, 1)$, then

$$Z^2 \sim \chi_{(1)}^2.$$

If $Z_1, Z_2, \dots, Z_p \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$, then

$$Z_1^2 + Z_2^2 + \dots + Z_p^2 = \sum_{i=1}^p Z_i^2 \sim \chi_{(p)}^2.$$

If $X_1, X_2, \dots, X_n$ are independently drawn from a $\mathcal{N}(\mu, \sigma^2)$ then

$$\begin{aligned}\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} &\sim \mathcal{N}(0, 1) \\ \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right)^2 &= \frac{(\bar{X} - \mu)^2}{\sigma^2/n} \sim \chi_{(1)}^2\end{aligned}$$

and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$, then

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2}.$$

3.1.2 t distribution

定义 3.2 t distribution

If two independent variables $Z \sim \mathcal{N}(0, 1)$ and $U \sim \chi_{(n)}^2$, then

$$t = \frac{Z}{\sqrt{U/n}} \sim t_{(n)}.$$

If $X_1, X_2, \dots, X_n$ are independently drawn from a $\mathcal{N}(\mu, \sigma^2)$, and we have got $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$ and $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2$, then

$$\frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} / (n-1)}} = \boxed{\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}}.$$

3.1.3 F distribution

定义 3.3 F distribution

If two independent variables $X \sim \chi_{(n)}^2$ and $Y \sim \chi_{(m)}^2$, then

$$F = \frac{X/n}{Y/m} \sim F_{(n,m)}.$$

Continuing from the definition of t -distribution:

$$t = \frac{Z}{\sqrt{U/n}} \sim t_{(n)} \quad \Rightarrow \quad t^2 = \frac{Z^2/1}{U/n} \sim F_{(1,n)}.$$

If $X_1, X_2, \dots, X_n$ are independently drawn from a $\mathcal{N}(\mu, \sigma^2)$, and we have got $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$, then

$$\left(\frac{\bar{X} - \mu}{S/\sqrt{n}} \right)^2 = \frac{(\bar{X} - \mu)^2}{S^2/n} \sim F_{(1,n-1)}.$$

3.1.4 Sampling distribution: population variance known vs unknown

σ^2 known

$$\begin{aligned} \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} &\sim \mathcal{N}(0, 1) \\ \frac{(\bar{X} - \mu)^2}{\sigma^2/n} &\sim \chi_{(1)}^2 \end{aligned}$$

σ^2 unknown (replaced with S^2)

$$\begin{aligned} \frac{\bar{X} - \mu}{S/\sqrt{n}} &\sim t_{(n-1)} \\ \frac{(\bar{X} - \mu)^2}{S^2/n} &\sim F_{(1,n-1)} \end{aligned}$$

3.2 Inferences about a mean vector

3.2.1 Hypothesis testing for μ

Suppose we are testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$. We reject H_0 if the statistic lies **outside** the critical values (two-tailed, the distribution is symmetric, e.g. $\mathcal{N}(\cdot, \cdot)$, $t_{(\cdot)}$), or **exceeds** the critical value (one-tailed, e.g. $\chi_{(\cdot)}^2$, $F_{(\cdot, \cdot)}$); otherwise, we do not reject H_0 .

Univariate

$$\begin{aligned} \frac{\bar{X} - \mu_0}{S/\sqrt{n}} &\sim t_{(n-1)} \\ \frac{(\bar{X} - \mu_0)^2}{S^2/n} &\sim F_{(1,n-1)} \end{aligned}$$

Multivariate (p -dimensional)

$$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim \frac{(n-1)p}{n-p} \cdot F_{(p, n-p)}$$

3.2.2 Simultaneous T^2 interval

We have known that

$$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim \frac{(n-1)p}{n-p} \cdot F_{(p, n-p)},$$

such that

$$\Pr(n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) < c^2) = 1 - \alpha,$$

where $c^2 = \frac{(n-1)p}{n-p} \cdot F_{(p, n-p)}(\alpha)$, then

$$\Pr\left((\bar{\mathbf{X}} - \boldsymbol{\mu})^\top (\bar{\mathbf{X}} - \boldsymbol{\mu}) < \frac{c^2 \mathbf{S}}{n}\right) = 1 - \alpha,$$

therefore, the interval of $\boldsymbol{\mu}_i$ is

$$\boxed{\bar{\mathbf{X}}_i \pm \sqrt{\frac{(n-1)p}{n-p} \cdot F_{(p, n-p)}(\alpha)} \times \sqrt{\frac{\mathbf{S}_{ii}}{n}}.$$

3.2.3 Bonferroni interval

Bonferroni method is similar with simultaneous T^2 method, that is

$$\boxed{\bar{\mathbf{X}}_i \pm t_{(n-1)}\left(\frac{\alpha}{2}\right) \times \sqrt{\frac{\mathbf{S}_{ii}}{n}}.$$

3.2.4 Large sample inference

Simultaneous

$$\bar{\mathbf{X}}_i \pm \sqrt{\chi_{(p)}^2(\alpha)} \times \sqrt{\frac{\mathbf{S}_{ii}}{n}}$$

Multivariate (p -dimensional)

$$\bar{\mathbf{X}}_i \pm Z\left(\frac{\alpha}{2p}\right) \times \sqrt{\frac{\mathbf{S}_{ii}}{n}}$$

4 Comparison of several multivariate means

4.1 Hotelling T^2 Test

4.1.1 Univariate review

Suppose,

$$X_1, \dots, X_n \sim \mathcal{N}(\mu_x, \sigma_x^2) \quad \text{and} \quad Y_1, \dots, Y_m \sim \mathcal{N}(\mu_y, \sigma_y^2).$$

The hypothesis test is $H_0 : \mu_x = \mu_y$ with the assumption of $\sigma_x = \sigma_y = \sigma$.

When σ is **known**,

$$\frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sigma \sqrt{\frac{1}{n} + \frac{1}{m}}} \sim \mathcal{N}(0, 1).$$

When σ is **unknown**, we replace it by pooling, that is

$$\frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}} \sim \mathcal{N}(0, 1),$$

$$\text{where } S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}.$$

4.1.2 Multivariate

定义 4.1 Hotelling T^2

If $\mathbf{X}_{11}, \mathbf{X}_{12}, \dots, \mathbf{X}_{1n_1}$, is a random sample of size n_1 , from $\mathcal{N}_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $\mathbf{X}_{21}, \mathbf{X}_{22}, \dots, \mathbf{X}_{2n_2}$ is an independent random sample of size n_2 from $\mathcal{N}_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$, then

$$T^2 = [(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p \right]^{-1} [(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)],$$

where

$$\mathbf{S}_p = \frac{(n_1-1)\mathbf{S}_1 + (n_2-1)\mathbf{S}_2}{n_1 + n_2 - 2},$$

is distributed

$$\frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - 1 - p} F_{(p, n_1 + n_2 - 1 - p)}.$$

4.2 $(1 - \alpha)\%$ confidence region for $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$

4.2.1 Contour equation

根据定义 2.2 【Contour equation】, 我们曾提到:

常密度轮廓的 p 维正态分布是定义为 $\mathbf{x}$ 的椭球体, 其方程为

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2 \sim \chi_{(p)}^2,$$

其中，椭球的中心 (center) 位于 $\boldsymbol{\mu}$ ，其两轴 (major/minor axis) 向量为 $\pm c\sqrt{\lambda_i}\mathbf{e}_i$ ，有

$$\boldsymbol{\Sigma}\mathbf{e}_i = \lambda_i\mathbf{e}_i,$$

即 λ_i 和 $\mathbf{e}_i$ 分别是 $\boldsymbol{\Sigma}$ 的特征值 (eigenvalue) 和特征向量 (eigenvector)，其中，两轴方向 (direction) 为 $\mathbf{e}_i$ ，两个半轴长度 (half-length) 为 $c\sqrt{\lambda_i}$ 。

那么在这里，critical value 为

$$c^2 = \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - 1 - p} F_{(p, n_1 + n_2 - 1 - p)},$$

方差 $\boldsymbol{\Sigma}$ 使用 pooled sample variance，即

$$\boldsymbol{\Sigma} = \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p,$$

但为了简便运算，我们只使用

$$\mathbf{S}_p\mathbf{e}_i = \lambda_i\mathbf{e}_i,$$

那这时

$$\boldsymbol{\Sigma}\mathbf{e}_i = \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p\mathbf{e}_i = \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \lambda_i\mathbf{e}_i,$$

故，半轴长度为

$$\sqrt{\lambda_i} \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) c^2}$$

4.2.2 Simultaneous confidence intervals

We have known that

$$[(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p \right]^{-1} [(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - 1 - p} F_{(p, n_1 + n_2 - 1 - p)},$$

such that

$$\Pr \left([(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p \right]^{-1} [(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] < c^2 \right) = 1 - \alpha,$$

where $c^2 = \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - 1 - p} F_{(p, n_1 + n_2 - 1 - p)}(\alpha)$, then

$$\Pr \left([(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top [(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] < c^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p \right) = 1 - \alpha,$$

therefore, the interval of $\boldsymbol{\mu}_i$ is

$$\boxed{(\bar{\mathbf{X}}_{1i} - \bar{\mathbf{X}}_{2i}) \pm \sqrt{\frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - 1 - p} F_{(p, n_1 + n_2 - 1 - p)}(\alpha)} \times \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{p,ii}}.$$

4.2.3 Bonferroni interval

Bonferroni method is similar with simultaneous T^2 method, that is

$$\boxed{(\bar{\mathbf{X}}_{1i} - \bar{\mathbf{X}}_{2i}) \pm t_{n_1+n_2-2} \left(\frac{\alpha}{2p} \right) \times \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{p,ii}}.}$$

4.3 Two population situation when $\Sigma_1 \neq \Sigma_2$

Let the sample sizes be such that $n_1 - p$ and $n_2 - p$ are large. Then, an approximate $100(1 - \alpha)\%$ confidence ellipsoid for $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ is given by all $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ satisfying

$$[(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left(\frac{1}{n_1} \mathbf{S}_1 + \frac{1}{n_2} \mathbf{S}_2 \right)^{-1} [(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \leq \chi_{(p)}^2(\alpha).$$

5 Multivariate linear regression

5.1 Review of univariate multiple regression

Model

$$\underset{n \times 1}{\mathbf{y}} = \underset{n \times (r+1)}{\mathbf{X}} \underset{(r+1) \times 1}{\boldsymbol{\beta}} + \underset{n \times 1}{\boldsymbol{\epsilon}},$$

i.e.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1r} \\ 1 & x_{21} & x_{22} & \cdots & x_{2r} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nr} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_r \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

Properties

- $\mathbb{E}[\boldsymbol{\epsilon}] = \mathbf{0}$; and
- $\text{Cov}[\boldsymbol{\epsilon}] = \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top] = \sigma^2 \mathbf{I}$.

Least square estimation

- Objective function (minimization):

$$S(\boldsymbol{\beta}) = \boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

- Estimation of coefficients:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

The properties:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} \quad \text{and} \quad \text{Cov}[\hat{\boldsymbol{\beta}}] = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

- Estimation of residuals:

$$\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \hat{\mathbf{y}} = \left[\mathbf{I} - \underbrace{\mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top}_{\text{hat matrix}} \right] \mathbf{y} = (\mathbf{I} - \mathbf{H}) \mathbf{y}.$$

The properties:

$$\mathbb{E}[\hat{\boldsymbol{\epsilon}}] = \mathbf{0} \quad \text{and} \quad \text{Cov}[\hat{\boldsymbol{\epsilon}}] = \sigma^2 (\mathbf{I} - \mathbf{H}).$$

5.2 Multivariate multiple regression

Model

$$\underset{n \times m}{\mathbf{Y}} = \underset{n \times (r+1)}{\mathbf{X}} \underset{(r+1) \times m}{\boldsymbol{\beta}} + \underset{n \times m}{\boldsymbol{\epsilon}},$$

where

$$\begin{aligned}\mathbf{Y}_{n \times m} &= \begin{bmatrix} Y_{11} & Y_{12} & \cdots & Y_{1m} \\ Y_{21} & Y_{22} & \cdots & Y_{2m} \\ \vdots & \vdots & & \vdots \\ Y_{n1} & Y_{n2} & \cdots & Y_{nm} \end{bmatrix} = \begin{bmatrix} \mathbf{Y}_{(1)} & \mathbf{Y}_{(2)} & \cdots & \mathbf{Y}_{(m)} \end{bmatrix}, \\ \mathbf{X}_{n \times (r+1)} &= \begin{bmatrix} x_{10} & x_{11} & x_{12} & \cdots & x_{1r} \\ x_{20} & x_{21} & x_{22} & \cdots & x_{2r} \\ \vdots & \vdots & \vdots & & \vdots \\ x_{n0} & x_{n1} & x_{n2} & \cdots & x_{nr} \end{bmatrix}, \\ \boldsymbol{\beta}_{(r+1) \times m} &= \begin{bmatrix} \beta_{01} & \beta_{02} & \cdots & \beta_{0m} \\ \beta_{11} & \beta_{12} & \cdots & \beta_{1m} \\ \beta_{21} & \beta_{22} & \cdots & \beta_{2m} \\ \vdots & \vdots & & \vdots \\ \beta_{r1} & \beta_{r2} & \cdots & \beta_{rm} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\beta}_{(1)} & \boldsymbol{\beta}_{(2)} & \cdots & \boldsymbol{\beta}_{(m)} \end{bmatrix}, \\ \boldsymbol{\epsilon}_{n \times m} &= \begin{bmatrix} \epsilon_{11} & \epsilon_{12} & \cdots & \epsilon_{1m} \\ \epsilon_{21} & \epsilon_{22} & \cdots & \epsilon_{2m} \\ \vdots & \vdots & & \vdots \\ \epsilon_{n1} & \epsilon_{n2} & \cdots & \epsilon_{nm} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\epsilon}_{(1)} & \boldsymbol{\epsilon}_{(2)} & \cdots & \boldsymbol{\epsilon}_{(m)} \end{bmatrix},\end{aligned}$$

with

$$\mathbb{E}[\boldsymbol{\epsilon}] = \mathbf{0} \quad \text{and} \quad \text{Cov}[\boldsymbol{\epsilon}_{(i)}, \boldsymbol{\epsilon}_{(k)}] = \sigma_{ik}^2 \mathbf{I}.$$

因此，我们可以简化成若干 univariate linear regression model，即

$$\mathbf{Y}_{(i)} = \mathbf{X} \boldsymbol{\beta}_{(i)} + \boldsymbol{\epsilon}_{(i)}, \quad i = 1, 2, \dots, m,$$

with $\text{Cov}[\boldsymbol{\epsilon}_{(i)}] = \sigma_{ii}^2 \mathbf{I}$.

5.3 Analysis of variance (ANOVA)

理论内容在此不过多赘述. 我们仅需清楚 ANOVA 是用来比较两个模型的手段即可，即在 Model 1 的基础上增加若干变量得到 Model 2 后是否更优.

$$\text{Model 1: } Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_r X_r,$$

$$\text{Model 2: } Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_r X_r + \beta_{r+1} X_{r+1} + \cdots + \beta_s X_s.$$

此时，假设检验为：

$$H_0 : \begin{pmatrix} \beta_{r+1} & \beta_{r+2} & \cdots & \beta_s \end{pmatrix}^\top = \mathbf{0} \text{ (每个变量都没有作用); vs.}$$

$$H_1 : \begin{pmatrix} \beta_{r+1} & \beta_{r+2} & \cdots & \beta_s \end{pmatrix}^\top \neq \mathbf{0} \text{ (至少一个变量有作用).}$$

ANOVA 的 R code 为 `anova(model_1, model_2)`，若其 p-value $< \alpha$ ，则拒绝 H_0 ，即在控制了 $X_1, \dots, X_r$ 的影响之后，解释变量 $X_{r+1}, \dots, X_s$ 对响应变量 Y 仍然具有显著的线性影响；也就是说，把 $X_{r+1}, \dots, X_s$ 加入模型可以显著提高对 Y 的拟合.

6 Data mining and machine learning

本章内容主要围绕着如何对若干相关变量进行“捏合”，比如数据集有从 0 岁到 90 岁每个年龄段的数据，但是将这 91 个变量同时放进模型中肯定是不可取的，原因有下：

1. 会极大增加模型的复杂度；
2. 模型假设 explanation variable 之间不应有过大的相关性（或称变量间应独立）；
3. ...

因此，提取这些相关变量的主要信息至关重要。

6.1 Principal component analysis (PCA)

6.1.1 Review of eigenvalues and eigenvectors

定理 6.1 Recall: eigenvalues and eigenvectors

For a symmetric matrix Σ , if the eigenvalues are $\lambda_1, \lambda_2, \dots, \lambda_p$, with the corresponding normalized eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_p$, then

$$\mathbf{e}_i^\top \mathbf{e}_i = 1,$$

and

$$\mathbf{e}_i^\top \mathbf{e}_j = 0, \quad \text{for } i \neq j.$$

In addition,

$$\sum_{i=1}^p \lambda_i = \text{trace}(\Sigma).$$

定理 6.2 Recall: diagonalised

Based on the definition of eigenvalues and eigenvectors, i.e.

$$\Sigma \mathbf{e}_i = \lambda_i \mathbf{e}_i,$$

we can get

$$\underset{(p \times p)}{\Sigma} = \sum_{i=1}^p \lambda_i \mathbf{e}_i \mathbf{e}_i^\top = \underset{(p \times p)}{\mathbf{P}} \underset{(p \times p)}{\Lambda} \underset{(p \times p)}{\mathbf{P}^\top},$$

where the normalised eigenvectors are the columns of another matrix

$$\mathbf{P} = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_p \end{bmatrix},$$

and $\mathbf{P}^\top \mathbf{P} = \mathbf{P} \mathbf{P}^\top = \mathbf{I}$, and Λ is a diagonal matrix

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_p \end{bmatrix}.$$

6.1.2 Principal component

定义 6.1 i th principal component

Let Σ be the covariance matrix associated with the random vector $\mathbf{X}^\top = [X_1 \ X_2 \ \cdots \ X_p]$. Let Σ have the eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \dots, (\lambda_p, \mathbf{e}_p)$. Then the i th **principal component** is given by

$$Y_i = \mathbf{e}_i^\top \mathbf{X} = e_{i1}X_1 + e_{i2}X_2 + \cdots + e_{ip}X_p, \quad i = 1, 2, \dots, p.$$

With these choices,

$$\begin{aligned} \text{Var}[Y_i] &= \mathbf{e}_i^\top \Sigma \mathbf{e}_i = \lambda_i, \quad i = 1, 2, \dots, p, \\ \text{Cov}[Y_i, Y_j] &= \mathbf{e}_i^\top \Sigma \mathbf{e}_j = 0, \quad i \neq j. \end{aligned}$$

定理 6.3 Total variation remains same

Let $\mathbf{X}^\top = [X_1 \ X_2 \ \cdots \ X_p]$. Let Σ have covariance matrix Σ , with eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \dots, (\lambda_p, \mathbf{e}_p)$. Let $Y_1 = \mathbf{e}_1^\top \mathbf{X}, Y_2 = \mathbf{e}_2^\top \mathbf{X}, \dots, Y_p = \mathbf{e}_p^\top \mathbf{X}$ be the principal components. Then

$$\sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} = \sum_{i=1}^p \text{Var}[X_i] = \lambda_1 + \lambda_2 + \cdots + \lambda_p = \sum_{i=1}^p \text{Var}[Y_i].$$

Proof.

$$\text{trace}(AB) = \text{trace}(BA).$$

The above theorem means the **total variation** can be calculated by

$$\lambda_1 + \lambda_2 + \cdots + \lambda_p,$$

then the **proportion of total variation** due to (explained by) the k th principal component is

$$\frac{\lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_p}, \quad k = 1, 2, \dots, p,$$

where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$ in general.

If most (for instance, 80% to 90%) of the total population variance, for large p , can be attributed to the first one, two, or three components, then these components can “replace” the original p variables without much loss of information.

6.2 Factor analysis

定义 6.2 Orthogonal factor model (正交因子模型)

$$\underset{(p \times 1)}{\mathbf{X}} - \underset{(p \times 1)}{\boldsymbol{\mu}} = \underset{(p \times m)}{\mathbf{L}} \underset{(m \times 1)}{\mathbf{F}^\top} + \underset{(p \times 1)}{\boldsymbol{\epsilon}^\top},$$

where

- $\mathbf{X}$: observable random vector, with p components;
- μ_i : mean of variable X_i ;
- ℓ_{ij} : loading of the i th variable on the j th factor;
- F_j : j th common factor;
- ϵ_i : i th specific factor.

The unobservable random vectors $\mathbf{F}$ and $\boldsymbol{\epsilon}$ satisfy the following conditions:

- $\mathbf{F}$ and $\boldsymbol{\epsilon}$ are independent;
- $\mathbb{E}[\mathbf{F}] = \mathbf{0}$, $\text{Cov}[\mathbf{F}] = \mathbf{I}$;
- $\mathbb{E}[\boldsymbol{\epsilon}] = \mathbf{0}$, $\text{Cov}[\boldsymbol{\epsilon}] = \boldsymbol{\Psi}$, where $\boldsymbol{\Psi}$ is a diagonal matrix.

Therefore, the covariance matrix of $\mathbf{X}$

$$\boldsymbol{\Sigma} = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi},$$

where the diagonal elements

$$\underbrace{\sigma_{ii}}_{\text{Var}[X_i]} = \underbrace{\ell_{i1}^2 + \ell_{i2}^2 + \cdots + \ell_{im}^2}_{\text{communality}} + \underbrace{\psi_i}_{\text{specific variance}},$$

we can write

$$\sigma_{ii} = h_i^2 + \psi_i, \quad \text{where } h_i^2 = \ell_{i1}^2 + \ell_{i2}^2 + \cdots + \ell_{im}^2.$$

In the PCA, we have $\boldsymbol{\Sigma} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^\top$, then $\mathbf{L} = \mathbf{P}\boldsymbol{\Lambda}^{\frac{1}{2}}$ and $\boldsymbol{\Psi} = \mathbf{0}$, hence the i th

6.3 Discrimination and classification

记符号 $c(a|b)$ 为将 b 判定为 a 的个数, 则 $\Pr(a|b)$ 为将 b 判定为 a 的概率.

定理 6.4 Expected cost of misclassification

将 p_1, p_2 分别记为两类的先验概率 (prior probability of population), 则

$$\text{ECM} = c(2|1) \Pr(2|1)p_1 + c(1|2) \Pr(1|2)p_2.$$

我们可以认为 ECM 是 “误判数 $\times$ 误判率 $\times$ 被误判出现的概率” 之和.

定理 6.5 Classification rule based on ECM

比较 “density ratio” 和 “cost ratio $\times$ prior probability ratio”, 即比较

$$\frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \quad \text{and} \quad \frac{c(2|1)}{c(1|2)} \times \frac{p_1}{p_2},$$

- 若 $\text{LHS} \geq \text{RHS}$, 则分类为 Region 1;
- 若 $\text{LHS} < \text{RHS}$, 则分类为 Region 2.

定义 6.3 Mahalanobis distance (马氏距离)

The square of Mahalanobis distance is defined as

$$d_k^2(\mathbf{x}) = (\mathbf{x} - \bar{\mathbf{x}}_k)^\top \mathbf{S}_p^{-1} (\mathbf{x} - \bar{\mathbf{x}}_k),$$

where pooled variance $\mathbf{S}_p = \frac{(n_1 - 1)\mathbf{S}_1 + (n_2 - 1)\mathbf{S}_2}{n_1 + n_2 - 2}$.

因此新参与者在马氏距离意义上更靠近哪个总体, 即比较 d_i^2 和 d_j^2 , 则可以判断新参与者的类别判定.

6.4 Cluster analysis

定义 6.4 Complete linkage (完全链接)

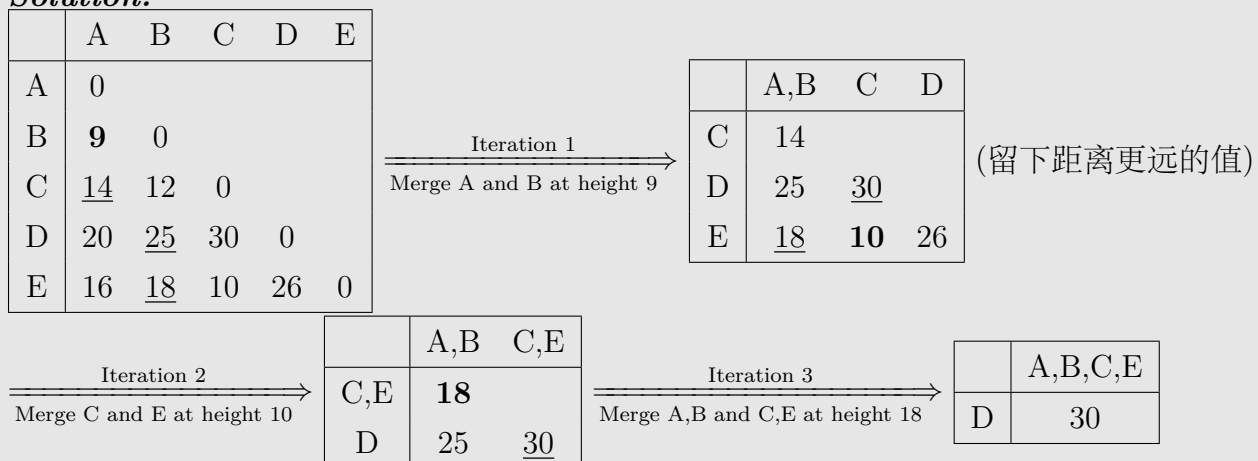
计算两个簇之间所有样本点的最大距离:

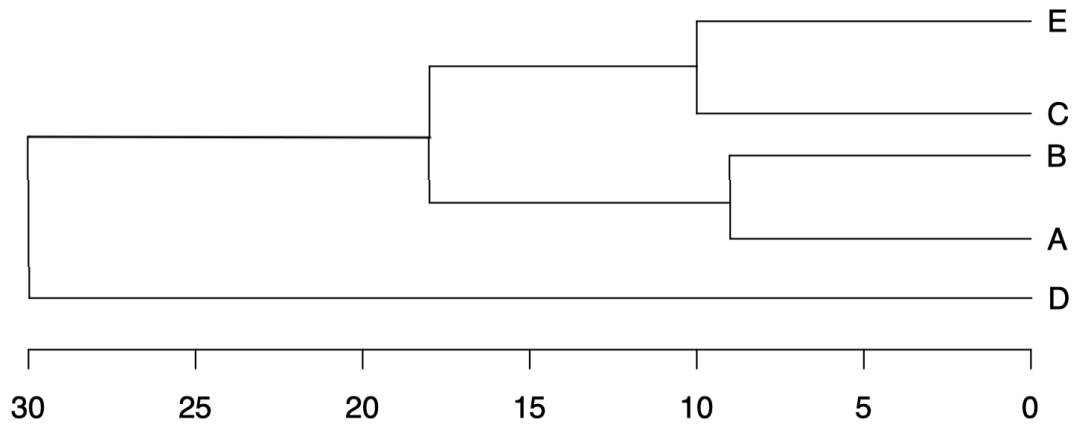
$$d(A, B) = \max_{i \in A, j \in B} d(i, j)$$

例 6.1

Draw the dendrogram to show the result of hierarchical agglomerative clustering using complete linkage. 绘制层次凝聚聚类树状图, 使用完全链接.

Solution.





分类 (classification) 与聚类 (clustering) 的根本区别 分类使用已有标签的训练样本来建立判别规则（监督学习）；而聚类是在没有类别标签的情况下，仅根据样本间相似性把数据自动分组（无监督学习）。

分类 (classification):

- 是监督学习 (supervised learning): 有明确的对应关系, 即 (x_i, y_i) , 需要明确 input 与 output;
- 目标: 学习一个判别规则, 来预测新样本的类别;
- 类别是**预先给定的**.

分类 (clustering):

- 是无监督学习 (unsupervised learning): 无明确的对应关系, 只有标签 x_i , 可认为是数据挖掘;
- 目标: 根据样本之间的“相似度/距离”自动把样本分成若干组;
- 类别 (簇) 是由算法**自动发现**的, 不是事先定义好的.